

4.1.1 Data, Information and Database

ڈیٹا خام حقائق ہیں، معلومات وہی حقائق ہیں جب انہیں ترتیب دے کر معنی دے دیا جائے۔

CS XI

Chapter 4: Data and Analysis

Lecture 01 of 15

4.1.1 Data, Information and Database

LEARN IT

Data are unprocessed facts, figures or observations. They can be **quantitative** (numbers: 80, 75, 90) or **qualitative** (descriptions: the colours of cars in a parking lot). Data on its own is not very useful — a list of marks does not yet tell you whether the class is strong.

When you organise, process and interpret data, it becomes **information**. Information has meaning and supports a decision. Example: "The class average is 80%, so we can move to the next chapter."

A **database** is an organised collection of **related** data, stored in a structured way so it can be accessed, managed and updated easily. Paper registers and loose files make searching painful; a database is built for search, update and sharing.

Data → Processing → Information → Decision

Aspect	Data	Information
Definition	Raw facts and figures	Processed, meaningful data
Example	85, 90, 75	Average marks = 83.3%
Usefulness	Little, on its own	Useful for decisions

WORKED EXAMPLE

Five CS XI marks: 80, 75, 90, 70, 85. Sum = 400. Average = $400 \div 5 = 80\%$. The numbers are data; "the class average is 80%" is information; a **Student** table holding every student's marks is a database.

REMEMBER IT

Exam line: *Data is raw; information is processed data used for decision-making. Always add one pair of examples.*

COMMON MISTAKE

Do not call a jumble of WhatsApp numbers a database. Related + structured + easy to update is what makes it a database.

اردو میں سمجھیں

ڈیٹا خام حقائق ہیں، معلومات وہی حقائق ہیں جب انہیں ترتیب دے کر معنی دے دیا جائے۔ ڈیٹا بیس متعلقہ ڈیٹا کا منظم ذخیرہ ہے جسے آسانی سے تلاش اور تبدیل کیا جا سکتا ہے۔

CHECK YOURSELF

1. Is "Nadia, Class 11, 72 marks" data or information?
2. Why is a stack of unmarked answer sheets not a database?

Answers: 1. Data (raw facts about one student). 2. It is not organised in a structured, searchable, updatable collection of related records.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Data is:

- a. Raw unprocessed facts
- b. Always information
- c. Always a graph
- d. A program

MCQ 2. Information is:

- a. Processed, meaningful data
- b. Noise only
- c. A cable
- d. A gate

MCQ 3. A database is:

- a. An organised collection of related data
- b. A single unsorted scrap
- c. A CPU fan
- d. A pie chart

MCQ 4. Marks 45, 67, 12 are:

- a. Data
- b. Information already
- c. A query
- d. A report

MCQ 5. "Ali failed" after processing marks is:

- a. Information
- b. Raw data only
- c. A primary key
- d. A form

MCQ 6. Storing related tables together is the job of a:

- a. Database
- b. Compiler
- c. Router
- d. K-map

Extra short questions

1. Differentiate data and information with a school example.
2. Define database.
3. Give two problems of storing data only in paper lists.
4. Is a phone contacts app a database?

Extra long question

1. A college keeps loose pages of marks. Argue, with definitions, why they should move to a database, and what information a principal could then get.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Data: 45, 67. Information: class average is 56 so we need extra classes.
2. Organised related data stored so it can be retrieved quickly.
3. Slow search, duplication, hard updates, errors.
4. Yes — organised related records (name, number).

Long

1. Define data/information/database. Benefits: search, consistency, reports (fail list, subject averages). One risk: need backup/access control.

Next lecture: 4.1.2 Database Management System (DBMS)